

Midterm 1

● Graded

Student

ANGAD KAPOOR

Total Points

98.5 / 100 pts

Question 1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 1.5 pts wrong $\sin 3x$ term
- 1.5 pts wrong $\cos 3x$ term
- 1.5 pts wrong e^{-6x} term
- 1.5 pts Missed x^2 for e^{-6x}
- 2 pts Messed up the algebraic term
- 2 pts missed x^2 for the algebraic term

Question 2

(no title)

8.5 / 10 pts

- 0 pts Correct

- 1 pt Missing a term for the polynomial part of the solution for $-7x^3e^{2x}$ (each term worth 1 point)
- 1.5 pts Wrong answer for the polynomial part of the particular solution for $-12xe^{4x}$

✓ - 1.5 pts Need to multiply the particular solution for $-12xe^{4x}$ with an extra x due to multiplicity.

- 1 pt Minor mistakes
- 2 pts Small conceptual error/ multiple minor mistakes

💬 check your roots, they are incorrect.

Question 3

(no title)

20 / 20 pts

✓ - 0 pts Correct

- 1.5 pts Missing multiplication by x for the e^{-2x} term for the particular solution.
- 1.5 pts some calculation/minor errors
- 3 pts Didn't solve for one of the particular coefficients/solved it wrong (each coeff worth 3 points)
- 6 pts Missing the form for y_p
- 6 pts Both coefficients for y_p are wrong/not calculated
- 2 pts Didn't put the homogenous and particular solution together.

Question 4

(no title)

20 / 20 pts

4.1 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 1 pt not getting $(-1)^3$
- 1 pt Not getting $\frac{d^3}{dp^3}$
- 1 pt Minor calculation mistakes
- 5 pts Fully wrong.
- 2 pts not going from $(-1)^3 \frac{d^3}{dp^3} e^{-2p}$ to $8e^{-2p}$

4.2 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 1.5 pts Wrong e^x part.
- 1 pt Messing up the cosh part.
- 1.5 pts wrong shifting formula for the cosh part
- 5 pts Wrong solution/not attempted
- 1 pt Minor mistake

4.3 (no title)

10 / 10 pts

✓ - 0 pts Correct

- 2.5 pts Significant error in one term plus in 'linearity'
- 3 pts Extra Term and error in rewriting $f(x)$ correctly
- 0.5 pts Small sign error
- 3 pts Missing a term and error in rewriting $f(x)$
- 1.5 pts error in exponent/denominator on one term
- 3 pts Error in integration computation
- 2 pts Significant error in one term
- 4 pts Missing a term and Significant error in rewriting $f(x)$
- 2.5 pts Major error in rewriting $f(x)$
- 4.5 pts Major errors in each term when applying L.T.
- 4 pts Error in integration for one term and evaluating at limits of integration
- 7 pts Error in rewriting $f(x)$ and L.T. never applied
- 4 pts Rewrote $f(x)$ with delta instead of U

Question 5

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 4 pts For not writing the integral form

- 3 pts Didn't compute integral

- 3 pts Didn't compute limits of integral

+ 1 pt for correct answer

- 3 pts partial work

- 1.5 pts Minor errors

Question 6

(no title)

15 / 15 pts

6.1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 10 pts Blank/No Meaningful Progress

- 8 pts Attempted to use the shift function.

- 6 pts Incorrect a value and missing sine term

- 1 pt arithmetic error

- 5.5 pts Missing inverse L.T. of e^{-p} and lost track of $1/2$'s

- 9 pts Major error

- 2 pts Small Error

6.2

(no title)

5 / 5 pts

- 0 pts Correct: FYI Using convolution would've been quicker

✓ - 0 pts Correct

- 0.5 pts One term has p in it

- 5 pts Incorrect

- 1 pt Error in PFD set-up

- 0.5 pts Error in PFD coefficient values

- 3.5 pts Set up PFD but then ran into trouble

Question 7

(no title)

15 / 15 pts

✓ + 15 pts Correct

- + 8 pts Applied L.T. correctly to the left side
- + 5 pts Applied L.T. correctly to right side
- + 2 pts Solved for $\mathcal{L}[y(x)]$
- + 6 pts Applied L.T. on left side correctly except didn't distribute negative for $-y'$
- + 2 pts Computed $\mathcal{L}[y'(x)]$ correctly but didn't distribute the negative
- + 0 pts Note that the table (and book) gives a generic formula for the L.T. of the n^{th} derivative of a function.
- + 2.5 pts all terms after the first should be subtracted when applying L.T. to the derivative of a function
- + 4 pts Computed L.T. of $y'(x)$ correctly
- + 2.5 pts Computed L.T. of $(-5x^2)$ correctly
- + 1.5 pts Stated L.T. of left hand side.
- + 6 pts Wrong ICs used
- + 1 pt Solved for $Y(p)$ but with error
- + 3 pts Unknown error while taking L.T. of left hand side
- + 5 pts L.T. of left side has error in L.T. of $y'(x)$

Midterm 1B

Name: ANGAD KAPOORStudent ID: 005-697-193

By signing below, you affirm that you have neither given nor received unauthorized help on this exam.

Signature: Angad**Instructions:**

- Do not open this exam until instructed to do so.
- You will have 50 minutes to complete the exam.
- This exam is worth a total of 100 points.
- Please print your name and student ID number above AND ON EVERY PAGE.
- **You may not use calculators, books, notes, or any other material to help you.**
- Please make sure **your phone and smartwatch is silenced** and stowed at the front of the room along with all bags/backpacks.
- You may use any available space on the exam for scratch work.
- **Answer must be written in the box provided!** Show your work on the rest of the page.
- You must **show your work** to receive credit.
- **Proper Notation and Organized.** Clear, concise, and correct use of mathematical notation is part of a correct answer. Work should be organized in a reasonable neat and coherent way in the space provided. Work scattered all over the page without a clearing ordering is not acceptable.

Good Luck!

1. (10 points) Determine the form of the particular solution of

$$y^{(4)} + 12y^{(3)} + 36y'' = e^{-3x} \sin(3x) + 5e^{-6x} + 8x$$

$$y_p = e^{-3x}(A \cos(3x) + B \sin(3x)) + Cx^2 e^{-6x} + x^2(Dx + F)$$

$$y_h: y^{(4)} + 12y^{(3)} + 36y'' = 0$$

$$\Rightarrow r^4 + 12r^3 + 36r^2 = 0$$

$$r^2(r^2 + 12r + 36) = 0$$

$$r^2(r+6)^2 = 0$$

$$\Rightarrow y_h = c_1 + c_2 x + c_3 e^{-6x} + c_4 x e^{-6x}$$

$$1 \rightarrow e^{-3x} \sin(3x) \Rightarrow e^{-3x}(A \cos(3x) + B \sin(3x))$$

$$1 \rightarrow 5e^{-6x} \Rightarrow Cx^2 e^{-6x} \quad (x^0, x^1 \text{ in } y_h)$$

$$1 \rightarrow 8x \Rightarrow x^2(Dx + F) \quad (x^2, x^1 \text{ in } y_h)$$

2. (10 points) Determine the form of the particular solution of

$$y'' - 16y = -7x^3 e^{2x} + 3 \cos(4x) - 12x e^{4x}$$

$$y_p = (Ax^3 + Bx^2 + Cx + D)e^{2x} + (F \cos(4x) + G \sin(4x)) + (Hx + J)e^{4x}$$

$$y_h: y'' - 16y = 0$$

$$\Rightarrow r^2 - 16r = 0$$

$$r(r-16) = 0$$

$$r = 0, r = 16$$

$$\Rightarrow y_h(x) = c_1 + c_2 e^{16x}$$

$$1 \rightarrow -7x^3 e^{2x} \Rightarrow (Ax^3 + Bx^2 + Cx + D)e^{2x}$$

$$1 \rightarrow 3 \cos(4x) \Rightarrow F \cos(4x) + G \sin(4x)$$

$$1 \rightarrow -12x e^{4x} \Rightarrow (Hx + J)e^{4x}$$

3. (20 points) Solve **completely** $y'' - 4y' - 12y = 4e^{3x} - 7e^{-2x}$ using Method of Undetermined Coefficients.

$$y(x) = c_1 e^{-2x} + c_2 e^{6x} - \frac{4}{15} e^{3x} + \frac{7}{8} x e^{-2x}$$

$$y_h = y'' - 4y' - 12y = 0$$

$$r^2 - 4r - 12 = 0$$

$$(r+2)(r-6) = 0$$

$$\Rightarrow y_h = c_1 e^{-2x} + c_2 e^{6x}$$

$$y_p = A e^{3x} + B x e^{-2x} \quad (\text{already in } y_h)$$

$$\therefore y = c_1 e^{-2x} + c_2 e^{6x} + A e^{3x} + B x e^{-2x}$$

$$\Rightarrow y' = -2c_1 e^{-2x} + 6c_2 e^{6x} + 3A e^{3x} + B e^{-2x} - 2B x e^{-2x}$$

$$\Rightarrow y'' = 4c_1 e^{-2x} + 36c_2 e^{6x} + 9A e^{3x} - 2B e^{-2x} - 2B e^{-2x} + 4B x e^{-2x}$$

$$= 4c_1 e^{-2x} + 36c_2 e^{6x} + 9A e^{3x} - 4B e^{-2x} + 4B x e^{-2x}$$

$$\therefore y'' - 4y' - 12y = (4c_1 e^{-2x} + 36c_2 e^{6x} + 9A e^{3x} - 4B e^{-2x} + 4B x e^{-2x})$$

$$- 4(-2c_1 e^{-2x} + 6c_2 e^{6x} + 3A e^{3x} + B e^{-2x} - 2B x e^{-2x})$$

$$- 12(c_1 e^{-2x} + c_2 e^{6x} + A e^{3x} + B x e^{-2x})$$

$$= (9A - 12A - 12A) e^{3x} + (-4B - 4B) e^{-2x} + (4B + 8B - 12B) x e^{-2x}$$

$$= -15A e^{3x} - 8B e^{-2x} = 4e^{3x} - 7e^{-2x}$$

$$\Rightarrow A = -\frac{4}{15}, \quad B = \frac{7}{8}$$

4. Find the Laplace transform of the following functions, if it exists. If it does not exist, justify. Your answer should not have any derivatives and/or integrals left in them.

(a) (5 points) $\mathcal{L}[x^3 \delta(x-4)]$

$$\mathcal{L}[x^3 \delta(x-4)] = 64 e^{-4p}$$

We know: $\mathcal{L}[(-1)^n x^n f(x)] = \frac{d^n}{dp^n} F(p)$ ($F(p) = \mathcal{L}[f(x)]$)

$$\Rightarrow \mathcal{L}[x^3 \delta(x-4)] = - \frac{d^3}{dp^3} \mathcal{L}[\delta(x-4)]$$

$$= - \frac{d^3}{dp^3} (e^{-4p})$$

$$= 4 \frac{d^2}{dp^2} (e^{-4p})$$

$$= -16 \frac{d}{dp} (e^{-4p})$$

$$= 64 e^{-4p}$$

(b) (5 points) $\mathcal{L}[e^x - 5e^{-3x} \cosh(2x)]$

$$= \frac{1}{p-1} - \frac{5p+15}{p^2+6p+5}$$

$$= \mathcal{L}[e^x] - 5 \mathcal{L}[e^{-3x} \cosh(2x)]$$

$$= \frac{1}{p-1} - 5 \left(\frac{(p+3)}{(p+3)^2 - 2^2} \right) \quad \dots \text{by Shifting Theorem}$$

$$= \frac{1}{p-1} - \frac{5p+15}{p^2+6p+5}$$

(c) (10 points) $\mathcal{L}[f(x)]$, where $f(x) = \begin{cases} 0 & 0 \leq x < 2 \\ 3 & 2 \leq x < 6 \\ e^{2x} & x \geq 6 \end{cases}$

$$= \frac{e^{-6p+12}}{p-2} + 3 \left(\frac{e^{-2p} - e^{-6p}}{p} \right)$$

By 'Activating' each part of the Heaviside function:

$$f(x) = e^{2x} u(x-6) + 3u(x-2) - 3u(x-6)$$

$$\begin{aligned} \Rightarrow \mathcal{L}[f(x)] &= \frac{e^{-6(p-2)}}{p-2} + \frac{3e^{-2p}}{p} - \frac{3e^{-6p}}{p} \\ &= \frac{e^{-6p+12}}{p-2} + 3 \left(\frac{e^{-2p} - e^{-6p}}{p} \right) \end{aligned}$$

5. (10 points) Use the **definition** to compute the Laplace transform of $f(x) = 3x - 7$

$$= \frac{3}{p^2} - \frac{7}{p}$$

$$\begin{aligned} \mathcal{L}[f(x)] &= \int_0^{\infty} e^{-px} f(x) dx = F(p) \\ \Rightarrow \mathcal{L}[3x-7] &= \int_0^{\infty} e^{-px} (3x-7) dx \\ &= 3 \int_0^{\infty} x e^{-px} dx - 7 \int_0^{\infty} e^{-px} dx \\ &= 3 \left(-\frac{x e^{-px}}{p} \Big|_0^{\infty} + \frac{1}{p} \int_0^{\infty} e^{-px} dx \right) - 7 \left(-\frac{1}{p} e^{-px} \Big|_0^{\infty} \right) \\ &= 3 \left(0 - \frac{1}{p^2} (e^{-px} \Big|_0^{\infty}) \right) - 7 \left(0 + \frac{1}{p} \right) = \frac{3}{p^2} - \frac{7}{p} \end{aligned}$$

6. Find the inverse Laplace transform of the following functions, if it exists. If the answer involves a convolution, you need to set up the integral but do not need to compute the integral. If it does not exist, justify.

(a) (10 points) $\mathcal{L}^{-1} \left[\frac{e^{-p}(1+p)}{2(p^2+1)} \right]$

$$\mathcal{L}[f(x-\alpha)u(x-\alpha)] = e^{-\alpha p} F(p)$$

$$\alpha=1, F(p)=\sin(x)$$

$$\Rightarrow \sin(x-1)u(x-1)$$

$$= \frac{1}{2} u(x-1) (\sin(x-1) + \cos(x-1))$$

$$= \mathcal{L}^{-1}[F(p)] \Rightarrow F(p) = \frac{e^{-p}}{2} \left(\frac{1}{p^2+1} + \frac{p}{p^2+1} \right) = \frac{1}{2} \left[e^{-p} \cdot \frac{1}{p^2+1} + e^{-p} \cdot \frac{p}{p^2+1} \right]$$

$$\Rightarrow = \frac{1}{2} [\sin(x-1)u(x-1) + \cos(x-1)u(x-1)]$$

$$= \frac{1}{2} u(x-1) (\sin(x-1) + \cos(x-1))$$

(b) (5 points) $\mathcal{L}^{-1} \left[\frac{1}{p^2(p-5)} \right]$

$$= \int_0^x e^{5\tau} (x-\tau) d\tau$$

$$= \mathcal{L}^{-1} \left[\frac{1}{p^2} \times \frac{1}{p-5} \right] = \mathcal{L}^{-1}[F(p) \cdot G(p)],$$

where $F(p) = \mathcal{L}[f(x)]$ and $G(p) = \mathcal{L}[g(x)]$

Then, by the Convolution Theorem.

$$\mathcal{L}^{-1} \left[\frac{1}{p^2} \times \frac{1}{p-5} \right] = \mathcal{L}^{-1} \left[\frac{1}{p^2} \right] * \mathcal{L}^{-1} \left[\frac{1}{p-5} \right]$$

$$= x * e^{5x}$$

$$\Rightarrow f(x) = \int_0^x e^{5\tau} (x-\tau) d\tau$$

7. (15 points) Begin solving the following IVP. Leave your answer in Laplace space, that is, leave your answer as $Y(p) =$

$$\begin{cases} y''' - y' = 2 - 5x^2 \\ y(0) = y'(0) = y''(0) = 1 \end{cases}$$

$$Y(p) = \frac{p^5 + p^4 + 2p^2 - 10}{p^3(p^3 - p)}$$

let $\mathcal{L}[y] = Y = Y(p)$

Applying Laplace to both sides:

$$\mathcal{L}[y''' - y'] = \mathcal{L}[2 - 5x^2]$$

$$\Rightarrow (p^3 \mathcal{L}[y] - p^2 y(0) - p y'(0) - y''(0)) - (p \mathcal{L}[y] - y(0)) = \frac{2}{p} - \frac{10}{p^3}$$

$$\Rightarrow (p^3 - p)Y - p^2 - p + 1 = \frac{2p^2 - 10}{p^3}$$

$$\begin{aligned} (p^3 - p)Y &= \frac{2p^2 - 10}{p^3} + (p^2 + p) = \frac{2p^2 - 10 + p^5 + p^4}{p^3} \\ &= \frac{p^5 + p^4 + 2p^2 - 10}{p^3} \end{aligned}$$

$$\Rightarrow Y(p) = \frac{p^5 + p^4 + 2p^2 - 10}{p^3(p^3 - p)}$$

